

# Supporting Information: Technical Details, Ablations, and Extended Validation for Deployment-Oriented Cooperative Yaw Control

June 19, 2026

This Supporting Information provides the full technical audit trail for the six core claims of the main paper: (1) the calibrated gray-box wake model is a trustworthy control-oriented surrogate; (2) the DRL policy recovers SLSQP-grade static cooperative-yaw performance; (3) the DRL policy does not depend on access to the SLSQP oracle; (4) the dynamic-wind degradation is linked to wake-propagation delay and partial observability; (5) DRL-Deploy—the policy plus supervisory wrapper—is the evaluated controller; (6) AEP, safety, and load-proxy results support deployment potential but require field and aeroelastic validation.

All dynamic comparisons use **percentage farm-power gain** and **perturbance yaw travel** under matched trajectories. Training rewards and regret-reward units are not used as deployment metrics.

## 1 Evaluation Protocols and Metric Definitions

### 1.1 Protocol overview

All evaluation protocols share the  $3\times 3$  NREL-5MW layout at  $7d_0$  spacing, a control period of  $T=10$ s, and the Config-E gray-box wake model as the evaluation environment. Table S1 summarises the protocols used throughout the paper.

### 1.2 Metric definitions

All metrics are reported per evaluation protocol. Table S2 defines the units and conventions.

All dynamic comparisons use the same wind trajectory realisations across controllers, ensuring matched evaluation conditions. Regret-reward units (used during training) are never reported as deployment metrics.

Table 1: Evaluation protocols. All dynamic protocols use AR(1)-generated wind trajectories matched across controllers.

| Protocol             | Wind model                                 | Conditions / trajectories | Steps                 |            |
|----------------------|--------------------------------------------|---------------------------|-----------------------|------------|
| Static evaluation    | Fixed $(\phi, v)$                          | 500 conditions            | Settle to convergence | DRL vs     |
| Dynamic Proto A      | AR(1), $\rho=0.99$ , $\sigma_\phi=1^\circ$ | 200 trajectories          | 200                   | Slow-var   |
| Dynamic Proto B      | AR(1), $\rho=0.95$ , $\sigma_\phi=2^\circ$ | 200 trajectories          | 200                   | Strong-var |
| Horns Rev AEP replay | i.i.d. von Mises + Weibull                 | 5,000 conditions          | 10 settling           | Annual     |
| Safety stress test   | Mixed wind (see S10)                       | 500 conditions            | Scenario-specific     | I          |

Table 2: Metric definitions and units. All gain values are percentage farm-power gain relative to the zero-yaw baseline.

| Metric                           | Definition                                                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Farm-power gain [%]              | $(P_{\text{ctrl}} - P_0)/P_0 \times 100$ , where $P_0$ is zero-yaw baseline power                                           |
| Gain/step [%]                    | Mean per-step gain over a trajectory; the deployment-relevant unit                                                          |
| Aligned-cube (AC) gain [%]       | Mean gain over conditions with $ \phi - 270^\circ  < 15^\circ$ and $v < 11.4$ m/s (matching the DRL-Deploy gate-entry band) |
| Yaw travel $[\circ/\text{turb}]$ | Cumulative absolute yaw change per turbine, summed over a trajectory                                                        |
| Peak yaw rate $[\circ/\text{s}]$ | Maximum per-step $ \Delta\gamma /T$ over a trajectory                                                                       |
| Negative-gain fraction [%]       | Fraction of control steps where $P_{\text{ctrl}} < P_0$                                                                     |
| Annual travel [Mdeg/yr]          | Total yaw travel extrapolated to 8,760 h at $T=10$ s                                                                        |
| Gain per travel                  | Mean gain [%] $\div$ mean travel per condition $[\circ]$                                                                    |
| Unsafe command count             | Steps where $ \gamma_i  > 30^\circ$ for any turbine OR $ \Delta\gamma /T > 5^\circ/\text{s}$                                |
| Fallback activation rate [%]     | Fraction of steps where gate output is zero (zero-yaw command)                                                              |

## 2 Gray-Box Wake-Model Calibration and Cross-Validation

### 2.1 CFD reference datasets and calibration

Three CFD reference datasets were used for joint parameter calibration:

- Three-turbine inline layout: wake-velocity profiles at  $d_0$  spacing;
- 35-turbine large farm at  $240^\circ$  inflow: farm-scale wake accumulation;
- Two-turbine yaw-sweep: yaw-induced deflection at  $\gamma \in [0^\circ, 50^\circ]$ .

The Bastankhah–Porté-Agel Gaussian wake model was calibrated via Levenberg–Marquardt least-squares optimisation over all three datasets jointly. The free

parameters are the wake-spreading coefficients ( $\alpha^*$ ,  $\beta^*$ ), the superposition mixing factor ( $\alpha$ ), and the ambient turbulence intensity ( $I$ ). Table S3 reports the calibrated values.

Table 3: Calibrated gray-box wake-model parameters.  $I$  is treated as a free calibration parameter (not a measured ambient TI) because the analytical model does not separately resolve wake-added turbulence.

| Parameter  | Calibrated value    | Initial value (Bastankhah–Porté-Agel 2014) |
|------------|---------------------|--------------------------------------------|
| $\alpha^*$ | 2.728               | 2.32                                       |
| $\beta^*$  | 0.100               | 0.154                                      |
| $\alpha$   | 0.540               | 1.00                                       |
| $I$        | 0.065 (lower bound) | 0.10                                       |

$I$  converges to its prescribed lower bound (0.065), indicating that the default turbulence parameterisation in the analytical model already captures the aggregate wake spreading at the reference turbine spacing adequately, and that the optimisation compensates for residual bias primarily through the spreading and mixing coefficients.

## 2.2 Horns Rev I validation

The calibrated model is validated against the Horns Rev I LES benchmark (Barthelmie et al., 2009) using the V80 turbine power curve (calibration datasets and validation figures are provided in the main text, Section 3). Normalised farm power as a function of wind direction is compared against LES measurements, confirming that the model captures the direction-dependent wake-loss structure without site-specific retuning.

## 2.3 FLORIS cross-validation

FLORIS cross-validation compares the calibrated gray-box model against FLORIS (GCH wake model, NREL 5MW turbine) on the  $3 \times 3$  layout for a direction sweep ( $\gamma=0^\circ$ ) and a two-turbine yaw sweep ( $\gamma \in [0^\circ, 50^\circ]$ ). The two models agree on the sign, rank ordering, and location of peak cooperative-yaw gains, supporting the use of the gray-box model as a DRL training surrogate.

# 3 PPO Training and Implementation Details

## 3.1 Observation and action spaces

The observation vector for history length  $J$  is the flattened concatenation of  $J$  observation rows. Each row contains: per-turbine yaw angles ( $N$ ), per-turbine inflow speeds ( $N$ ), wind-direction cosine and sine, wind speed (3), downstream-lock mask ( $N$ ), and—with position encoding enabled—normalised  $(x, y)$  turbine coordinates ( $2N$ ). For  $N=9$  turbines with position encoding, one row has

$3N+3+2N=48$  dimensions. The total observation dimension is  $J \times 48$  (Table S4). The action is an incremental yaw command  $\Delta\gamma_i \in [-AB, +AB]$  per turbine, accumulated and clipped to  $[-50^\circ, +50^\circ]$ .

Table 4: Observation dimensions for the three history lengths evaluated.

| $J$ | Obs. dimension | Temporal coverage at $T=10$ s |
|-----|----------------|-------------------------------|
| 3   | 144            | 30 s                          |
| 8   | 384            | 80 s                          |
| 15  | 720            | 150 s                         |

### 3.2 Hyperparameters and checkpoint selection

The Config-E PPO hyperparameters are listed in Table S5. The Lower Confidence Bound (LCB) criterion is used for checkpoint selection:  $\mu - 1.96\sigma/\sqrt{n_{\text{seeds}}}$  over the final 10 training iterations, applied to the episode return. All five training seeds use different random initialisations (seed 0–4) and identical hyperparameters.

Table 5: PPO hyperparameters (Config-E).

| Hyperparameter          | Value                                                    |
|-------------------------|----------------------------------------------------------|
| Policy network          | MLP [128, 128], ReLU                                     |
| Value network           | MLP [128, 128], ReLU                                     |
| Discount $\gamma$       | 0.995                                                    |
| GAE $\lambda$           | 0.98                                                     |
| Learning rate           | $3 \times 10^{-4} \rightarrow 3 \times 10^{-5}$ (cosine) |
| Batch size              | 4,096                                                    |
| Epochs per update       | 10                                                       |
| Clip range              | 0.2                                                      |
| Entropy coefficient     | 0.01                                                     |
| Max KL (early stop)     | 0.015                                                    |
| Optimiser               | AdamW, weight decay $2 \times 10^{-4}$                   |
| Parallel environments   | 128                                                      |
| Total environment steps | $6 \times 10^7$                                          |
| Control period $T$      | 10 s                                                     |
| Action bounds           | $\pm 10^\circ$ per step                                  |
| Observation history $J$ | 3                                                        |

### 3.3 JAX/NNX on-device implementation

The training loop is fully on-device: the wake model, the PPO rollout (128 parallel environments), and the GAE+PPO update are all expressed in JAX and JIT-compiled. This achieves  $\sim 95,000$  FPS on an NVIDIA RTX 4090 (see main

text, Section 3, for the JAX throughput benchmark). The JAX environment produces numerically equivalent results to the original NumPy gym on a 19-point  $(\phi, v)$  grid, confirming that the acceleration does not alter the physical evaluation model.

## 4 Configuration Ablations Leading to Config-E

### 4.1 Wind-sampling mixture

The wind-sampling distribution during training affects which inflow regimes the policy experiences. Table S6 reports the steady-state aligned-cube gain for four mixtures under the baseline PPO configuration ( $\pm 5^\circ$  bounds,  $3 \times 10^7$  steps, 3 seeds). The balanced mixture (aligned 0.3, near-aligned 0.3, off-axis 0.4) is adopted in Config-E as a compromise between aligned-cube concentration and sufficient exposure to off-axis conditions.

Table 6: Steady-state aligned-cube gain vs. training wind-sampling mixture (baseline Config-D,  $\pm 5^\circ$ , 3 seeds,  $3 \times 10^7$  steps).

| Mixture (aligned/near/off-axis) | AC gain [%] |
|---------------------------------|-------------|
| Uniform (0/0/1)                 | +4.16       |
| Focused (0.5/0.3/0.2)           | +4.78       |
| Balanced (0.3/0.3/0.4)          | +4.89       |
| Aggressive (0.7/0.2/0.1)        | +4.84       |

### 4.2 Action-bound sensitivity

Expanding the action bounds from  $\pm 5^\circ$  to  $\pm 10^\circ$  per step allows the policy to accumulate yaw more rapidly during the settling phase of each condition. Under the focused sampling mixture and  $6 \times 10^7$ -step budget, the  $\pm 10^\circ$  configuration yields an aligned-cube gain of +5.61% (vs. +4.89% at  $\pm 5^\circ$ , Table S6, Balanced), at the cost of increased dynamic-wind yaw travel (from  $52^\circ$  to  $210^\circ/\text{turb}$ ).

### 4.3 Downstream-lock ablation

The downstream-turbine locking mechanism prevents turbines that are not significantly waking any downstream neighbour from participating in cooperative yaw. To verify that locking remains essential under the optimized settings, we re-trained the **Config-E controller** with the lock-mask disabled (lock\_off\_configE) for three random seeds ( $3 \times 10^7$  steps, Config-E hyperparameters:  $J=3$ , deficit normalization, position encoding, SLSQP-regret reward, focused sampling 0.3/0.3/0.4,  $\pm 10^\circ$  action bounds, cosine LR, AdamW). The lock-enabled comparison uses the Config-E **sens\_act10** checkpoints trained under identical settings.

The lock-enabled policy achieves a mean regret-reward gain of +2.42% over 200 uniformly sampled wind conditions, whereas the lock-disabled policy degrades to −0.46% with an increased negative-gain fraction (28% vs. 18%; see main text, Section 3.2). Disabling the lock erases the cooperative gain and destabilises learning even under the optimized Config-E settings: rear turbines, lacking downstream peers, yaw away from the incoming flow under stochastic exploration, paying a  $\cos^{1.88}(\gamma)$  power tax for no recoverable benefit. The locking mechanism is thus a necessary structural ingredient of the framework, validated under Config-E and retained in all configurations.

#### 4.4 KL early-stopping

KL-targeted early stopping functions as a safety valve rather than a primary regularisation mechanism. Table S7 reports the final training metrics for two Config-D variants (3 seeds,  $3 \times 10^7$  steps, TARGET\_KL=0.015 enabled vs. disabled via a high threshold). Cosine learning-rate decay and AdamW weight decay are the dominant contributors to Config-D’s improved training stability.

Table 7: KL early-stop ablation: final training metrics (3-seed mean  $\pm$  std, Config-D settings).

| Metric                          | KL ON (0.015)     | KL OFF (100)       |
|---------------------------------|-------------------|--------------------|
| Final episode reward            | +344.0 $\pm$ 93.7 | +311.0 $\pm$ 138.6 |
| Mean KL divergence (last iter.) | 0.0021            | 0.0021             |
| Early-stop trigger rate         | 0.3%              | 0.0%               |

#### 4.5 Reward-penalty ablation

Adding yaw-magnitude and yaw-rate penalties ( $\lambda_{\text{mag}} = \lambda_{\text{rate}} = 8 \times 10^{-5}$ ) to the training reward depresses the final-10-iteration return from +19.79 to +15.78 (−20% relative) and lowers the best-seed peak from +28.1 to +22.3 (see main text, Section 4.1). This result informed the decision to introduce rate penalties only through dynamic-wind training ( $\lambda_{\text{rate}}$ , Section S6) rather than via static-training reward shaping.

### 5 Reward and Algorithmic Baselines

#### 5.1 Oracle-free marginal-reward ablation

The SLSQP-regret reward uses a precomputed SLSQP lookup table ( $91 \times 11$  grid) to provide a headroom signal during training. To test whether the DRL policy’s cooperative-yaw capability depends on this signal, we trained a Config-E variant with a marginal reward (power gain relative to zero-yaw baseline,

Table 8: SLSQP-regret vs. oracle-free marginal reward: per-seed results (Config-E settings,  $6 \times 10^7$  steps).

| Reward               | AC gain [%]      | Dynamic gain/step [%] | Travel/turb [ $^\circ$ ] |
|----------------------|------------------|-----------------------|--------------------------|
| SLSQP-regret         | $+4.48 \pm 6.17$ | $-0.158$              | 210                      |
| Marginal (no oracle) | $+4.13 \pm 5.56$ | $-0.070$              | 115                      |

no SLSQP reference). Table S8 reports the full seed-wise comparison (3 seeds,  $6 \times 10^7$  steps, Config-E settings).

The marginal-reward policy reaches 92.2% of the SLSQP-regret aligned-cube gain, confirming that the DRL policy does not depend on the SLSQP oracle for its cooperative-yaw capability. The SLSQP-regret formulation acts as a sample-efficiency accelerator—it raises the static peak—while the oracle-free policy exhibits lower dynamic-wind travel, consistent with a more conservative action profile. These data are reported as 3-seed means from the reward-ablation experiment; full seed-wise data are available in `results/reward_ablation.json`.

## 5.2 Behavioral Cloning baseline

A Behavioral Cloning (BC) baseline was trained via supervised regression to predict SLSQP-optimal yaw vectors from wind conditions, using a dataset of 1,000 SLSQP solutions sampled from the training distribution. The BC policy uses the same [128, 128] MLP architecture as the PPO actor and was trained for 100 epochs with MSE loss.

Evaluated on 100 uniformly sampled wind conditions, the BC policy achieves a marginal mean gain of  $-20.20\%$  and an aligned-cube gain of  $-31.50\%$  ( $n=7$  conditions)—substantially worse than the zero-yaw baseline. The result suggests that direct supervised regression to SLSQP yaw vectors is insufficient under the present dataset size and architecture, likely because the SLSQP yaw mapping contains discontinuities near regime boundaries. The DRL policy, by contrast, learns through interaction and produces smoother yaw trajectories. The BC result is reported as an auxiliary negative baseline; it is not a central claim of the paper.

## 5.3 SAC baseline

A JAX/NNX-native Soft Actor-Critic (SAC) agent was implemented on the same on-device stack. The SAC agent uses twin Q-networks with [128, 128] MLPs (identical to PPO), a Gaussian policy with tanh squashing, automatic entropy tuning, and a replay buffer of  $5 \times 10^5$  transitions. Training used  $N_{\text{envs}}=4$  parallel environments (vs. PPO’s 128; the lower count is a consequence of JIT compilation constraints for SAC’s multi-gradient update structure) and  $N_{\text{steps}}=512$  steps per rollout. Five seeds were trained for  $3 \times 10^7$  environment steps each under Config-E settings.

At the matched  $3 \times 10^7$ -step budget, SAC achieves a five-seed-mean final episode return of  $+185.2 \pm 75.3$  (regret-reward units), compared to PPO Config-E’s  $+334.3 \pm 217.0$ . SAC reaches 55.4% of PPO’s mean reward with substantially lower inter-seed variance. The throughput gap is  $13\times$  ( $\sim 7,100$  FPS for SAC vs.  $\sim 95,000$  FPS for PPO). We attribute SAC’s underperformance to off-policy sample inefficiency when the reward signal is concentrated in a narrow wind regime ( $\sim 9\%$  of conditions), and to the squashed-Gaussian action parameterisation penalising large yaw increments.

PPO is retained as the primary algorithm because it offers the best empirical performance-throughput compromise under the present implementation. SAC is not claimed to be unsuitable for wind-farm control in general; a pure-JAX refactoring of the gradient computation is expected to narrow the throughput gap.

## 6 Dynamic-Wind Benchmark: Full Results

### 6.1 Controller list and matched trajectory protocol

All dynamic-wind comparisons use the same AR(1) wind trajectory realisations. The seven controllers evaluated are:

1. Zero yaw (greedy, baseline);
2. Static Config-E (raw, no wrapper);
3. Dynamic  $\lambda_{\text{rate}}=0$  (no rate penalty, raw);
4. Dynamic  $\lambda_{\text{rate}}=5 \times 10^{-4}$  (raw);
5. Dynamic  $\lambda_{\text{rate}}=2 \times 10^{-3}$  (raw);
6. SLSQP lookup, rate-limited to  $0.5^\circ/\text{s}$ ;
7. SLSQP lookup, unlimited.

### 6.2 Protocol A and Protocol B results

Tables S9–S10 report the full unified dynamic-wind benchmark results (200 AR(1) trajectories, 200 steps each,  $T=10$  s). Proto A uses  $\rho=0.99$ ,  $\sigma_\phi=1^\circ$  (slow-varying wind); Proto B uses  $\rho=0.95$ ,  $\sigma_\phi=2^\circ$  (strong-variability wind). All DRL results are 5-seed means.

### 6.3 Historical Config-A actuator-aware experiment

The actuator-aware DRL framework was originally demonstrated with the baseline PPO configuration (CONFIG-A,  $3 \times 10^7$  steps,  $j=1$ , uniform sampling, marginal reward). Five DRL configurations were trained with varying yaw-rate penalties  $\lambda_{\text{rate}} \in \{0, 8 \times 10^{-5}, 5 \times 10^{-4}, 2 \times 10^{-3}, 10^{-2}\}$ , and compared against the SLSQP



Table 9: Unified dynamic-wind benchmark: Protocol A ( $\rho=0.99$ ,  $\sigma_\phi=1^\circ$ ). 5-seed mean DRL, single deterministic evaluation for SLSQP.

| Controller                             | Gain/step [%] | Travel/turb [ $^\circ$ ] | Peak rate [ $^\circ/\text{s}$ ] |
|----------------------------------------|---------------|--------------------------|---------------------------------|
| Zero yaw                               | 0.000         | 0                        | 0.00                            |
| Static Config-E (raw)                  | -0.163        | 165                      | 1.66                            |
| Dynamic $\lambda=0$                    | -1.932        | 456                      | 4.88                            |
| Dynamic $\lambda=5\times 10^{-4}$      | -0.870        | 393                      | 4.93                            |
| Dynamic $\lambda=2\times 10^{-3}$      | -1.054        | 376                      | 4.29                            |
| SLSQP lookup, RL=0.5 $^\circ/\text{s}$ | +2.178        | 277                      | 0.33                            |
| SLSQP lookup, unlimited                | +3.386        | 640                      | 4.73                            |

Table 10: Unified dynamic-wind benchmark: Protocol B ( $\rho=0.95$ ,  $\sigma_\phi=2^\circ$ ). DRL-Deploy controllers also shown.

| Controller                             | Gain/step [%] | Travel/turb [ $^\circ$ ] | Peak rate [ $^\circ/\text{s}$ ] |
|----------------------------------------|---------------|--------------------------|---------------------------------|
| Zero yaw                               | 0.000         | 0                        | 0.00                            |
| Static Config-E (raw)                  | -0.201        | 191                      | 0.21                            |
| DRL-Deploy (Static)                    | -0.083        | 16                       | 0.30                            |
| DRL-Deploy (J=15 Dyn)                  | +0.034        | 24                       | 0.30                            |
| SLSQP lookup, RL=0.5 $^\circ/\text{s}$ | +2.178        | 277                      | 0.33                            |
| SLSQP lookup, unlimited                | +3.386        | 640                      | 4.73                            |

lookup table under four rate-limit settings (unlimited, 0.5 $^\circ/\text{s}$ , 0.3 $^\circ/\text{s}$ , 0.1 $^\circ/\text{s}$ ) on 1000 AR(1) trajectories. The baseline DRL policies achieved gains of +0.11% to +0.24% with 116–129 $^\circ$  cumulative yaw travel and peak rates of 0.12–0.14 $^\circ/\text{s}$ —well within actuator limits. The unlimited lookup table achieved +1.90% gain but required 3.32 $^\circ/\text{s}$  peak rate and 2152 $^\circ$  travel; rate-limiting to 0.1 $^\circ/\text{s}$  reduced its gain to +1.04%. The rate-penalized DRL policies achieved comparable gain-per-travel efficiency (0.10–0.20 per 100 $^\circ$ ) to rate-limited lookup tables (0.09–0.21) with 4–8 $\times$  lower travel. These Config-A results motivated the systematic configuration optimization (Config-A  $\rightarrow$  Config-E) and the deployment-oriented controller development presented in the main text.

## 6.4 Interpretation

Raw static DRL and the SLSQP lookup table occupy opposite extremes of the power–travel trade-off. Raw DRL has low travel but negative mean gain; the unlimited lookup has high gain but 4.73 $^\circ/\text{s}$  per-turbine peak rate—far beyond the 0.3–0.5 $^\circ/\text{s}$  capability of utility-scale yaw drives. Dynamic-wind training without rate penalties over-reacts (2.4 $\times$  travel increase from static to  $\lambda=0$  under Proto A). Rate penalties mitigate the over-reaction but do not alone close the deployment gap: temporal context (J=15) and the supervisory wrapper (DRL-Deploy) are required to achieve positive per-step gain with actuator-compatible travel. The DRL-Deploy results in Table S10 (Protocol B) represent the final

deployment-oriented evaluation.

## 7 Extended Observation Horizon and Wake-Delay Hypothesis

### 7.1 Training setup

To test the wake-propagation-delay hypothesis—that a short observation window cannot perceive the causal effect of yaw actions on downstream turbines, causing spurious wind-action correlations—we trained Config-E policies with observation histories  $J=3, 8, 15$  under dynamic AR(1) wind. All policies use  $\lambda_{\text{rate}}=5 \times 10^{-4}$ ,  $6 \times 10^7$  training steps, and 5 seeds. The observation dimensions are given in Table S4.  $J=15$  (150 s at  $T=10$  s) spans the  $\sim 90$ –110 s wake-propagation delay at  $7d_0$  spacing and 8–10 m/s inflow.

### 7.2 Results

Table S11 reports the 5-seed dynamic-wind results under both protocols. Extending the observation window to  $J=15$  materially improves dynamic performance relative to  $J=3$  (Proto B:  $-0.534\%$  vs.  $-1.210\%$ ), while  $J=8$  is inconclusive and does not consistently improve over the short-window baseline (the  $J=8$  Proto A result is marginally worse than  $J=3$ ). The improvement is therefore concentrated at the longest observation horizon, consistent with the hypothesis that the observational window must substantially exceed the wake-propagation delay to provide useful temporal context.

Table 11: Dynamic-wind performance vs. observation history length  $J$  (5-seed mean,  $\lambda_{\text{rate}}=5 \times 10^{-4}$ ,  $6 \times 10^7$  steps).

| Configuration                      | $J$ | Proto A gain [%] | Proto B gain [%] | Travel/turb [°] |
|------------------------------------|-----|------------------|------------------|-----------------|
| Static $J=3$ (reference)           | 3   | $-0.178$         | $-0.120$         | 180             |
| Dynamic $\lambda=5 \times 10^{-4}$ | 3   | $-1.519$         | $-1.210$         | 376             |
| Dynamic $\lambda=5 \times 10^{-4}$ | 8   | $-1.667$         | $-1.231$         | 344             |
| Dynamic $\lambda=5 \times 10^{-4}$ | 15  | $-0.937$         | $-0.534$         | 273             |

### 7.3 Causal ablation

To distinguish genuine temporal-context exploitation from a mere dimensionality effect, we performed a causal ablation on the  $J=15$  observation buffer (single seed, 500 AR(1) trajectories, Proto B). Table S12 reports the results. Restricting to the last 3 frames collapses gain from  $+0.175\%$  to  $-0.201\%$ , confirming that information beyond the immediate 30 s history is essential. Shuffling the frame order degrades gain to  $-0.087\%$ , demonstrating that the policy relies on temporal ordering, not merely the presence of additional frames. Masking

frames  $t-10-t-12$  (100–130 s ago, the epoch closest to the wake-propagation delay) yields +0.159%—a modest degradation consistent with the partial loss of delay-scale information.

Table 12: Causal ablation of the J=15 observation buffer (single seed, 500 trajectories, Proto B).

| Observation mask | Gain/step [%] | Travel/turb [ $^{\circ}$ ] | Neg. frac. [%] |
|------------------|---------------|----------------------------|----------------|
| Full J=15        | +0.175        | 78.9                       | 15.0           |
| Last 3 only      | −0.201        | 91.1                       | 28.2           |
| Mask $t-10-t-12$ | +0.159        | 65.7                       | 14.5           |
| Shuffle          | −0.087        | 124.6                      | 16.7           |

## 7.4 Interpretation

The causal ablation provides evidence that the J=15 policy uses temporally ordered observational history beyond the most recent 30 s. The modest impact of masking the  $t-10-t-12$  band relative to the Last-3 collapse suggests that the policy draws on multiple time scales across the full 150 s window rather than relying predominantly on a single delay-epoch. These results validate the design rationale for extended observation windows but do not imply that temporal context alone solves dynamic control: the J=15 policy must still be combined with the DRL-Deploy supervisory wrapper to achieve deployment-relevant performance (Section S8, main text Section 4.8).

# 8 DRL-Deploy Wrapper Sensitivity

## 8.1 Wrapper definition

The DRL-Deploy controller is defined in Algorithm S1. The supervisory wrapper applies four sequential layers to the raw DRL policy output: (i) sector gate with hysteresis, (ii) yaw-command deadband, (iii) per-step rate limit, and (iv) zero-yaw fallback. The gate and fallback use only real-time wind measurements available to any SCADA system.

Table 13: DRL-Deploy wrapper parameters (nominal configuration).

| Parameter            | Value                                                             |
|----------------------|-------------------------------------------------------------------|
| Gate entry band      | $ \phi-270^{\circ}  < 15^{\circ}$                                 |
| Gate exit band       | $ \phi-270^{\circ}  < 20^{\circ}$ (hysteresis)                    |
| Gate speed ceiling   | $v < 11.4$ m/s                                                    |
| Yaw-command deadband | $2^{\circ}$ (per-turbine max $ \Delta\gamma $ )                   |
| Rate limit           | $\pm 3^{\circ}$ per control step (0.3°/s per turbine at $T=10$ s) |
| Fallback command     | $\gamma_i = 0^{\circ}$ (zero cooperative yaw)                     |

**Algorithm S1:** DRL-Deploy control cycle (one step,  $T=10$  s).

**Input:** wind direction  $\phi_t$ , wind speed  $v_t$ , previous yaw  $\gamma_{t-1}$ , gate state  $g_{t-1} \in \{0, 1\}$

**Output:** yaw command  $\gamma_t$ , gate state  $g_t$

1.  $d_\phi \leftarrow \min(|\phi_t - 270^\circ|, 360^\circ - |\phi_t - 270^\circ|)$
2.  $\theta \leftarrow 20^\circ$  if  $g_{t-1}=1$  else  $15^\circ$  // hysteresis
3.  $g_t \leftarrow (d_\phi < \theta) \wedge (v_t < 11.4 \text{ m/s})$
4. if  $g_t = 0$ : output  $\gamma_t \leftarrow 0^\circ$ ; return // fallback
5.  $\Delta\hat{\gamma} \leftarrow \pi_{\text{DRL}}(\text{obs}_t)$  // PPO forward pass
6. if  $\max_i |\Delta\hat{\gamma}_i| < 2^\circ$ :  $\Delta\hat{\gamma} \leftarrow 0$  // deadband
7.  $\Delta\gamma \leftarrow \text{clip}(\Delta\hat{\gamma}, -3^\circ, +3^\circ)$  // rate limit (0.3°/s)
8.  $\gamma_t \leftarrow \text{clip}(\gamma_{t-1} + \Delta\gamma, -50^\circ, +50^\circ)$
9. output  $\gamma_t, g_t$

## 8.2 Deadband sensitivity

The  $2^\circ$  deadband is tuned to the DRL policy’s characteristic action-increment distribution. Table S13 reports the dynamic-wind performance (Proto B, 5-seed mean) under deadband values  $\delta \in \{0^\circ, 1^\circ, 2^\circ, 3^\circ, 5^\circ\}$ .  $\delta=0^\circ$  (no deadband) gives the highest travel;  $\delta=5^\circ$  suppresses virtually all yaw activity because the policy’s typical increments ( $\sim 0.03^\circ/\text{step}$  in the dynamic-wind benchmark) lie far below the threshold.  $\delta=2^\circ$  is identified as the sweet spot for this application.

Table 14: Deadband sensitivity (Static DRL-Deploy, Proto B, 5-seed mean).

| Deadband $\delta$ | Gain/step [%] | Travel/turb $^\circ$ | Neg. frac. [%] |
|-------------------|---------------|----------------------|----------------|
| $0^\circ$         | -0.083        | 29.7                 | 19             |
| $1^\circ$         | -0.083        | 21.3                 | 19             |
| $2^\circ$         | -0.083        | 16.0                 | 18             |
| $3^\circ$         | -0.085        | 4.3                  | 20             |
| $5^\circ$         | -0.001        | 0.2                  | 1              |

## 8.3 Gate-threshold sensitivity

Table S14 reports sensitivity to the gate entry band ( $10^\circ, 15^\circ, 20^\circ$ ) with hysteresis exit at entry+ $5^\circ$ . Narrower gates reduce travel but may exclude genuine wake-aligned conditions; wider gates increase activation frequency but dilute the cooperative-yaw benefit with off-axis intervention.

## 8.4 Rate-limit verification

The DRL-Deploy controller applies a binding rate limit of  $\pm 3^\circ$  per control step ( $0.3^\circ/\text{s}$  per turbine at  $T=10$  s), enforced through the `RATE_MAX=0.3` parameter in the evaluation scripts. This is consistent with the  $0.3\text{--}0.5^\circ/\text{s}$  yaw-rate capability

of multi-MW utility-scale turbines. The per-turbine peak yaw rates of DRL-Deploy controllers (Tables S9–S10) are at this rate-limit bound.

## 9 Horns Rev AEP Replay Details

### 9.1 Wind-climate model

The Horns Rev I wind climatology is modelled as a Weibull wind-speed distribution ( $k=2.1$ ,  $A=10.5$  m/s at 70 m hub height) and a von Mises directional distribution ( $\mu=270^\circ$ ,  $\kappa=2.0$ ), based on published measurements (Barthelmie et al., 2009; Hansen et al., 2012). The baseline annual energy production is  $500 \text{ MW} \times 0.45 \text{ capacity factor} \times 8,760 \text{ h} = 1,971,000 \text{ MWh/yr}$ . The aligned-cube occurrence probability (i.i.d. sampling) is 9.1%, and the near-aligned probability ( $5^\circ < |\Delta\phi| \leq 15^\circ$ ) is 17.6%.

### 9.2 Monte Carlo sampling and annualisation

$n=5,000$  wind conditions are sampled i.i.d. from the joint distribution. Each condition is evaluated with 10-step settling (the DRL policy accumulates yaw through repeated incremental actions over the settling phase; the final-step power is used for the gain calculation). Annual yaw travel is annualised as  $\text{travel}_{\text{annual}} = \bar{t}_{\text{cond}} \times 8,760 \times 3,600 / (T \times N_{\text{settle}}) / 10^6 \text{ [Mdeg/yr]}$ , where  $\bar{t}_{\text{cond}}$  is the mean per-condition travel.

### 9.3 Controller comparison

Table S15 reports the full AEP replay results (main text Table 16).

Table 15: Horns Rev AEP replay: full controller comparison ( $n=5,000$  conditions, 10-step settling, 5 DRL seeds).

| Controller                      | AEP gain [%] | Annual travel [Mdeg] | Gain/travel | Peak rate [ $^\circ/\text{s}$ ] |
|---------------------------------|--------------|----------------------|-------------|---------------------------------|
| Greedy yaw                      | 0.000        | 0.000                | —           | 0.00                            |
| Static DRL-Deploy               | −0.694       | 8.169                | −0.027      | 0.30                            |
| J=15 DRL-Deploy                 | +0.122       | 2.098                | +0.018      | 0.30                            |
| SLSQP RL= 0.5 $^\circ/\text{s}$ | +0.455       | 4.572                | +0.031      | 0.44                            |
| SLSQP unlimited                 | +0.754       | 14.475               | +0.016      | 4.47                            |

### 9.4 Sensitivity to wind-direction concentration and bias

To verify that the AEP ranking (J=15 Deploy as the most deployment-favourable DRL variant) is robust to climatological uncertainty, we evaluate both DRL-Deploy controllers under varying wind-direction concentration  $\kappa \in [1.0, 8.0]$  and directional bias  $\pm 5^\circ$ . Table S16 reports the  $\kappa$  sensitivity; Table S17 reports the bias sensitivity. Static Deploy uses  $n=2,000$  conditions per parameter value;

J=15 Deploy uses  $n=500$  (the smaller sample reflects the higher per-condition evaluation cost of the J=15 policy, and values should be interpreted with correspondingly wider confidence intervals).

The J=15 Deploy AEP remains near-neutral to slightly positive across all  $\kappa$  and bias values tested, while Static Deploy is consistently negative. The qualitative ranking—J=15 Deploy outperforms Static Deploy under the Horns Rev wind climate—is preserved across the full sensitivity range, confirming that the main AEP finding is not an artifact of the nominal climatological parameters.

Table 16: AEP sensitivity to direction concentration  $\kappa$ : both DRL-Deploy controllers.

| $\kappa$ | AC prob. [%] | Static Deploy AEP [%] | J=15 Deploy AEP [%] | Static travel [Mdeg] |
|----------|--------------|-----------------------|---------------------|----------------------|
| 1.0      | 6.2          | −0.519                | −0.005              | 5.61                 |
| 2.0      | 9.1          | −0.672                | −0.014              | 7.65                 |
| 4.0      | 12.8         | −1.119                | +0.250              | 10.78                |
| 8.0      | 19.1         | −1.587                | +0.286              | 15.27                |

Table 17: AEP sensitivity to wind-direction bias: both DRL-Deploy controllers.

| Bias [°] | Static Deploy AEP [%] | J=15 Deploy AEP [%] | Static travel [Mdeg] |
|----------|-----------------------|---------------------|----------------------|
| −5       | −0.700                | −0.010              | 7.28                 |
| −2       | −0.550                | +0.177              | 7.35                 |
| 0        | −0.672                | +0.083              | 7.65                 |
| +2       | −0.702                | +0.186              | 7.72                 |
| +5       | −0.754                | +0.159              | 7.58                 |

## 9.5 Interpretation

J=15 DRL-Deploy is the only non-baseline controller achieving positive annualised gain with the lowest yaw travel. It does not maximise AEP—SLSQP achieves  $3.7\times$  higher AEP gain at  $2.2\times$  higher travel cost, and  $6.2\times$  higher AEP at  $6.9\times$  higher travel in the unlimited case—but it offers the most deployment-favourable power–travel compromise among the evaluated controllers. The sensitivity analysis confirms that this qualitative conclusion is robust to plausible uncertainty in the Horns Rev wind-direction climatology.

# 10 Safety and Fail-Safe Stress Tests

## 10.1 Fault definitions

Seven categories of abnormal conditions are tested (Table S18). Each scenario uses 500 mixed wind conditions (aligned, near-aligned, and cross-wind). The

unsafe-command threshold is  $|\gamma_i| > 30^\circ$  (any turbine) or  $|\Delta\gamma|/T > 5^\circ/\text{s}$  (any turbine).

Table 18: Safety-test fault definitions.

| Fault category      | Description                                       | Levels                                   |
|---------------------|---------------------------------------------------|------------------------------------------|
| Wind-direction bias | Constant offset added to $\phi$ observation       | $\pm 2^\circ, \pm 5^\circ, \pm 10^\circ$ |
| Wind-speed bias     | Constant offset added to $v$ observation          | $\pm 0.5, \pm 1.0 \text{ m/s}$           |
| Yaw-angle bias      | Constant offset added to $\gamma$ observation     | $\pm 2^\circ, \pm 5^\circ$               |
| Measurement dropout | Random packet loss with zero-order hold           | 5%, 10%                                  |
| Sudden wind shift   | Step change in $\phi$ at aligned inflow           | $+10^\circ, +20^\circ$                   |
| High-wind OOD       | Wind speeds above rated ( $\geq 17 \text{ m/s}$ ) | 17, 20, 25 m/s                           |
| Low-wind OOD        | Wind speeds below cut-in                          | 3, 4, 5 m/s                              |

## 10.2 Full scenario results

Table S19 reports the main safety metrics for all scenarios and controllers. The safety evaluation uses single-condition settling from a cold start (zero-filled observation history). Under this protocol, the J=15 DRL-Deploy policy’s rate-penalty-trained action increments are sufficiently small to be fully suppressed by the  $2^\circ$  deadband, producing  $\max |\gamma| = 0^\circ$  and zero unsafe commands—a worst-case conservative bound. Under trajectory-based evaluation with accumulated observation history (Sections S6–S7, main text Section 4.8), the J=15 policy produces non-zero yaw commands and achieves positive gains. The safety result therefore demonstrates that even when initialised without any temporal context, the J=15 Deploy controller defaults to safe (zero-yaw) behaviour rather than issuing unsafe commands.

## 10.3 Key findings

1. **Raw DRL is inherently unsafe:** 728 unsafe commands in clean conditions,  $45.5^\circ$  max yaw.
2. **DRL-Deploy wrapper provides substantial safety improvement:** unsafe commands reduced by 61% (Static) to 100% (J=15) relative to Raw DRL.
3. **Yaw-angle sensor bias is the most critical fault mode:**  $\pm 5^\circ$  bias produces 1,242–1,350 unsafe commands for Static Deploy, with max yaw reaching  $55^\circ$  (near the  $\pm 50^\circ$  physical limit). Yaw-encoder redundancy is recommended for field deployments.
4. **Gate provides automatic high-wind protection:** 100% fallback activation at  $v \geq 17 \text{ m/s}$ .

Table 19: Safety-test results: key metrics (single-seed, 500 conditions per scenario).

| Scenario          | Controller    | Gain/step [%] | Unsafe | Max $ \gamma $ [°] | Fallback [%] |
|-------------------|---------------|---------------|--------|--------------------|--------------|
| Clean             | Raw DRL       | −1.044        | 728    | 45.5               | 0            |
| Clean             | Static Deploy | −0.736        | 284    | 42.7               | 67           |
| Clean             | J=15 Deploy   | +0.000        | 0      | 0.0                | 67           |
| Yaw bias +5°      | Static Deploy | −3.080        | 1350   | 55.0               | 67           |
| Yaw bias +5°      | J=15 Deploy   | −2.613        | 890    | 50.2               | 67           |
| Yaw bias −5°      | Static Deploy | −2.936        | 1242   | 55.0               | 67           |
| Yaw bias −5°      | J=15 Deploy   | −1.810        | 832    | 50.0               | 67           |
| $\phi$ bias +10°  | Static Deploy | −0.480        | 203    | 42.7               | 74           |
| $\phi$ bias +10°  | J=15 Deploy   | +0.000        | 0      | 0.0                | 74           |
| $v$ bias +1.0 m/s | Static Deploy | −0.577        | 209    | 40.9               | 73           |
| Sudden shift +20° | Raw DRL       | +40.0→−21.7   | 40     | 38.5               | —            |
| Sudden shift +20° | Static Deploy | +40.0→0.0     | 0      | 26.3               | —            |
| Dropout 10%       | Static Deploy | −0.677        | 0      | 42.8               | 0            |
| OOD $v=17$ m/s    | Static Deploy | 0.000         | 0      | 0.0                | 100          |
| OOD $v=25$ m/s    | Static Deploy | 0.000         | 0      | 0.0                | 100          |

5. **Sudden wind shifts are handled gracefully:** DRL-Deploy returns to zero yaw within the gate exit band; Raw DRL exhibits catastrophic gain collapse (+40.0%→−21.7%) on a 20° shift.
6. **Measurement dropout (10%) induces negligible degradation.**

## 11 Load Proxy and Yaw-Activity Analysis

### 11.1 Yaw-activity proxies

Without full aeroelastic simulation (OpenFAST/FAST.Farm), we report two classes of first-order screening metrics.

**Yaw-activity proxies** (Table S20): cumulative yaw travel, peak yaw rate, direction reversals, cumulative squared yaw change ( $\sum \Delta\gamma^2$ ), and cumulative absolute yaw rate ( $\sum |\Delta\gamma|/\Delta t$ ). These quantify the mechanical duty on the yaw system.

**Aerodynamic side-load proxies** (Table S21, main text Limitations):  $\sum U_i^2 \gamma_i^2$  (proportional to yaw-misalignment aerodynamic moment) and  $\sum U_i^2 |\sin \gamma_i|$  (proportional to side force). These are computed from the unified dynamic-wind benchmark (Proto B, 5-seed Config-E mean) and are evaluated at each control step using the instantaneous inflow velocity and yaw angle.

Static training yields  $5.3\times$  lower  $\sum \Delta\gamma^2$  and  $2.6\times$  lower  $\sum |\Delta\gamma|/\Delta t$  compared to dynamic-wind training at  $\lambda_{\text{rate}}=5\times 10^{-4}$ , consistent with the observation that post-hoc gating provides a more favourable power-load trade-off than dynamic retraining. These are first-order approximations; they are not fatigue loads and should not be interpreted as structural design metrics.



Table 20: Yaw-activity proxies (5-seed Config-E mean, Proto B dynamic-wind benchmark).

| Controller                         | $\sum \Delta\gamma^2$ | $\sum  \Delta\gamma /\Delta t$ | Reversals |
|------------------------------------|-----------------------|--------------------------------|-----------|
| Zero yaw                           | 0                     | 0                              | 0         |
| Static Config-E                    | $2.54 \times 10^6$    | 4,007                          | 1,662     |
| Dynamic $\lambda=5 \times 10^{-4}$ | $1.35 \times 10^7$    | 10,419                         | 1,684     |

## 12 Scalability and Layout Generalization

### 12.1 $5 \times 5$ layout

Config-E policies were trained on a  $5 \times 5$  regular grid (25 turbines,  $7d_0$  spacing, 3 seeds,  $6 \times 10^7$  steps) with position encoding enabled. The  $5 \times 5$  aligned-cube gain is +4.70% ( $n=51$  conditions, 3-seed mean), transferring 95.7% of the  $3 \times 3$  headline (+4.91%). Without position encoding, the  $5 \times 5$  gain was +0.98%, confirming that position encoding is essential for larger layouts where the policy must distinguish between turbines at different spatial coordinates.

### 12.2 Irregular layout transfer

A Config-E policy trained on the regular  $3 \times 3$  grid was evaluated zero-shot on an irregular 9-turbine layout. The zero-shot aligned-cube gain is  $-1.41\%$  (negative, indicating the policy actively reduces farm power). Retraining the same Config-E architecture on the irregular layout from scratch recovers a positive aligned-cube gain of +1.39% (3 seeds,  $6 \times 10^7$  steps). A staggered  $3 \times 3$  layout (partial intermediate regularity) achieves +1.61% zero-shot transfer.

These results indicate that the DRL framework supports site-specific retraining but does not generalise zero-shot to arbitrarily irregular layouts. Position encoding provides partial transfer capability, but for deployment on a new layout, retraining with the target turbine coordinates is recommended.

## 13 Reproducibility Checklist

All result JSON files are committed in the `results/` directory. Trained policy checkpoints are available from the corresponding author upon reasonable request due to file-size constraints ( $\sim 200$  MB per seed).

Table 21: Reproducibility checklist. Script paths are relative to the `codes/` directory.

| Item                              | Value / Script                                                                                                    |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Code repository                   | <a href="https://github.com/VickyLastShao/wind-farm-yaw-rl">https://github.com/VickyLastShao/wind-farm-yaw-rl</a> |
| Commit hash (latest)              | 1da613a                                                                                                           |
| Python version                    | 3.13                                                                                                              |
| JAX version                       | 0.9.0.1                                                                                                           |
| Flax (NNX) version                | 0.12.4                                                                                                            |
| CUDA / GPU                        | CUDA 12.x, NVIDIA RTX 4090                                                                                        |
| FLORIS version                    | 4.x (GCH wake model)                                                                                              |
| SLSQP grid resolution             | $91 \times 11$ ( $\phi \in [173^\circ, 353^\circ]$ , $v \in [6, 16]$ m/s)                                         |
| Training random seeds             | 0–4                                                                                                               |
| Evaluation random seed            | 20260617 (dynamic), 42 (AEP), 123 (safety)                                                                        |
| Main text Table / Figure          | Reproducing Script                                                                                                |
| Table 2 (FLORIS cross-validation) | <code>eval_floris_configE.py</code>                                                                               |
| Table 8 (Closed-loop performance) | <code>eval_unified_static_vs_slsqp.py</code>                                                                      |
| Table 9 (Reward ablation)         | <code>run_ablation_marginal_reward.sh</code>                                                                      |
| Table 12 (Noise robustness)       | <code>eval_noise_batched.py</code>                                                                                |
| Table 14 (Gated DRL)              | <code>eval_gated_fast.py</code>                                                                                   |
| Table 15 (DRL-Deploy benchmark)   | <code>eval_deploy_final.py</code>                                                                                 |
| Table 16 (Horns Rev AEP)          | <code>eval_hornsrev_aep_deploy.py</code>                                                                          |
| Table 17 (J=15 causal ablation)   | <code>eval_j15_causal.py</code>                                                                                   |
| Safety tests (Section 4.8.2)      | <code>eval_safety_abnormal.py</code>                                                                              |
| Load proxy (Section 4.5)          | <code>compute_load_proxy.py</code>                                                                                |